

Emilio A. Magaña

Product Engineer

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Product Engineer with 2+ years of engineering experience and 1.5+ years building production web applications. Specializing in FPGA validation and TypeScript, ReactJS, NodeJS with proven ability to deliver scalable full-stack solutions. Proficient in AI-augmented development workflows using Claude/GitHub Copilot for generating boilerplate code, API scaffolding, type definitions, and accelerating full-stack implementation cycles. Strong foundation in systems engineering from FPGA validation work at Lattice Semiconductor.

EXPERIENCE

Product Engineer | Lattice Semiconductor

October 2021 – January 2024, Hillsboro, Oregon, USA

- Led validation and characterization of FPGA IP (I/O interfaces, PLLs, Oscillators) ensuring compliance with design specifications.
- Wrote comprehensive technical documentation for validation and characterization methodologies.
- Developed behavioral code for which to test IP under, using Verilog.
- Managed end-to-end bench operations from initial bring-up to full automatization using Python, ensuring reliability.
- Created data visualizations comparing specification targets against performance metrics using R, informing product decisions.
- Collaborated with cross-functional teams (sales, engineering, customers) to resolve technical issues via JIRA, translating complex technical concepts for diverse audiences.

Teacher's Assistant/Head TA | Oregon State University

January 2019 – March 2019, Corvallis, Oregon, USA

- Lead lab sessions for Introduction to Electrical and Computer Engineering.

PROJECTS

yourGuide | your-guide-flame.vercel.app

- Engineered a scalable tour booking application using the MERN stack (MongoDB, ExpressJS, ReactJS, NodeJS) with TypeScript.
- Applied secure role-based authentication, managing 3 user types (Admin, Guide, Basic User) with JWT and bcrypt password hashing.
- Built a RESTful API with 15+ protected endpoints, middleware authentication, and centralized error handling.
- Integrated Stripe payment processing for secure transactions.

Portfolio | magemi.dev

- Developed a responsive portfolio website using ReactJS and NextJS, styled with Tailwind CSS.
- Integrated custom RAG chatbot using OpenAI API, Astra DB vector storage and Upstash Redis caching for quick response times.
- Implemented server-side rendering for SEO optimization.
- Deployed on Vercel with automated CI/CD.

yourGetAway | yourgetaway.netlify.app/login

- Developed a full-stack property rental platform using ReactJS, replicating core Airbnb functionality.
- Implemented role-based access control, enabling authenticated users to manage reservations through a dashboard interface.
- Integrated Supabase for backend database management, handling user authentication, and data.

EDUCATION

Bachelor of Science in Electrical and Computer Engineering | Oregon State University

June 2021, Corvallis, Oregon, USA

- USA-GPA: 3.52/4.0 | DE-GPA: 1.63 (Gut).
- Minor in Computer Science.

SKILLS

Frontend: ReactJS, NextJS, AngularJS, TailwindCSS, HTML5, CSS3

Backend: NodeJS, NestJS, ExpressJS, MongoDB, RESTful APIs, Authentication (JWT), Supabase

Tools: OpenAI, Astra DB, Upstash Redis, Stripe API, Git, Vite, Figma

Languages: TypeScript, Python, Go, Verilog, R

Conversational Languages: English (Fluent), Spanish (Fluent), German (Conversational)